Code No: 155CK **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, September - 2021
NATURAL LANGUAGE PROCESSING
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks

1.a) List the applications and challenges in NLP.
b) Explain any one Morphological model. [7+8]

2.a) Discuss about challenging issues of Morphological models.
b) Differentiate between surface and deep structure in NLP with suitable examples. [7+8]

3.a) Explain various types of parsers in NLP.
b) Discuss multilingual issues in detail. [8+7]

4.a) Given Grammar
S→ AB | BB
A→ CC | AB | a
B→ BB | CA | b
C→ BA | AA | b
Word w= aabb. Apply Top Down Parsing test, the word can be generated or not.
b) Explain Tree Banks and its role in parsing. [8+7]

5.a) State the types of references in discourse.
b) Explain about Homonymy and Polysemy with suitable examples. [8+7]

6.a) How Morphological structure helps to Language Modelling?
b) How to handle semantics? [8+7]

7.a) Give example for Predicate Argument Structure.
b) What are System Paradigms in predicates? [7+8]

8.a) Consider the following training set and implement Bi-gram model to calculate the probability of given Test sentence.
Training set: She said thank you.
She said bye as she walked through the door.
She went to San Diego
Test sentence: She thanked and walk through the door
b) Give some examples for early NLP systems. [8+7]

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